

“NextGen Travel Intelligence and Automation Portal”

A Project Report Submitted in the partial fulfillment of the requirements for the award of the degree of

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In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled **“Design and Development of an Integrated Tourism Platform for Flights, Trains, Hotels, and Hidden Destinations”** is a record of Bonafide work of **ALLURU AMULYA - 2520030032 , CHILUMULA MADHULIKA - 2520090056**, submitted in

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CERTIFICATE

This is certify that the mini project based report entitled **“Design and Development of an Integrated Tourism Platform for Flights, Trains, Hotels, and Hidden Destinations”** is a bonafide work done and submitted by **ALLURU AMULYA - 2520030032 , CHILUMULA MADHULIKA - 2520090056**

in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering, K L (Deemed to be University), during the academic year **2024-2025.**

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Abstract

The NextGen Travel Intelligence and Automation Portal is designed to make corporate travel simpler, faster, and more organized. In many organizations, managing employee travel involves multiple steps like planning trips, booking flights, getting approvals, and tracking expenses, which can be time-consuming and complex. This system brings all these activities into one easy-to-use platform, helping companies handle travel more efficiently while reducing manual work and errors. The portal allows administrators to set clear travel rules such as budget limits, preferred airlines, and booking conditions, ensuring that every trip follows company policies. Employees can easily request travel through a simple interface, and managers can quickly approve or reject requests in real time. To make the system more advanced and practical, features like smart flight suggestions, price alerts, fraud detection, travel history, notifications, and role-based dashboards can be included. In addition, the system can offer options for in-flight food selection, provide detailed information about the pilot and crew, and allow users to compare different flights based on company policies, time, and cost. It can also include live flight status tracking, digital boarding passes for a paperless experience, and multi-language support to improve accessibility for users from different regions. Additional functionalities such as emergency support, payment integration, and mobile access make the system more flexible and user-friendly.

Finally, the proposed portal delivers a complete and efficient solution for managing corporate travel in a smart and organized manner. It not only helps in tracking travel expenses and maintaining all records in a centralized system, but also ensures better transparency and control over employee travel activities. The inclusion of user-friendly dashboards and basic analytics makes it easier for organizations to understand travel patterns and manage budgets effectively. Overall, this project aims to simplify complex travel processes by combining automation, policy control, and data management, resulting in improved efficiency, reduced costs, and a more convenient travel experience for both employees and management.

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CHAPTER -1 INTRODUCTION

1.1 Background of the Project

Corporate travel has become an essential component of modern business operations, enabling organizations to expand their reach, maintain client relationships, and participate in global markets. With the increasing frequency of employee travel, managing bookings, expenses, approvals, and tracking has become a complex and time-consuming task for companies. Traditionally, organizations relied on manual processes, spreadsheets, or third-party travel agencies to handle travel arrangements. These methods often lack efficiency, transparency, and real-time control.

With the advancement of information and communication technologies, digital solutions have transformed how travel is planned and managed. Today, employees can book flights, select seats, and manage itineraries through online platforms. However, despite these advancements, corporate travel management remains fragmented. Employees frequently use multiple systems for booking, expense tracking, and approvals, leading to duplication of data, inconsistencies, and reduced productivity.

The idea for this project emerged from the need to simplify and centralize corporate travel processes into a single, integrated platform. The proposed system, “Design and Development of a Corporate Travel Management Portal,” aims to provide a unified solution that streamlines travel booking, monitoring, and management. By integrating features such as flight selection, seat booking, baggage management, live flight tracking, and payment processing, the platform enhances user convenience and operational efficiency.

In addition to functionality, the project emphasizes user experience by providing an intuitive and interactive interface. Features such as live flight tracking with animated visualization, structured dashboards, and simplified workflows make the system accessible to users with varying levels of technical expertise. This ensures that both employees and administrators can efficiently manage travel activities without complexity.

From an organizational perspective, the system improves transparency and control over travel operations. It allows companies to monitor employee travel, enforce travel policies, and analyze expenses through centralized data. This not only reduces operational costs but also enhances accountability and decision-making.

Academically, the project integrates multiple core areas of computer science, including:

1. Web Development: Designing responsive and interactive user interfaces using modern frameworks.
2. Database Management: Storing and managing user data, bookings, and travel records efficiently.
3. API Integration: Connecting with external services for real-time flight and travel data.
4. Software Engineering: Implementing modular design, scalability, and maintainability.
5. User Experience Design: Ensuring usability, accessibility, and smooth navigation.

Furthermore, the system aligns with the growing trend of digital transformation in businesses, where automation and data-driven solutions are replacing manual processes. It supports organizations in adapting to modern travel requirements by providing a smart, efficient, and scalable platform.

To summarize, the background of this project is based on three key observations:

- Existing corporate travel processes are inefficient and fragmented.
- Technological advancements enable integration and automation of travel services.
- Organizations require better transparency, control, and user-friendly systems for managing employee travel.

By addressing these challenges, the project establishes a robust foundation for a centralized corporate travel management system that enhances efficiency, reduces costs, and improves the overall travel experience.

1.2 Problem Statement

In today's digital era, air travel booking systems are widely available, yet many of them fail to provide a complete and integrated experience for users, especially in a structured application environment. Most existing systems focus only on basic ticket booking and lack additional functionalities such as real-time tracking, seat management, baggage selection, and centralized control. This creates an incomplete and less efficient user experience.

One of the major problems addressed by this project is the lack of an all-in-one flight booking platform that combines multiple essential features into a single system. In many existing applications, users can book tickets but are unable to perform related tasks such as selecting seats, managing baggage, or visualizing their journey in a simplified and interactive way. This leads to a disconnected workflow and reduces user convenience.

Another key issue is the absence of real-time interaction within the system. Traditional booking platforms do not provide dynamic features like live flight tracking or visual movement representation. As a result, users do not get a clear understanding of their journey flow, making the system feel static and less engaging.

Seat management is also a challenge in basic systems. Without a proper seat locking mechanism, multiple users may attempt to select the same seat simultaneously, leading to conflicts and poor user experience. Similarly, baggage selection is often not integrated into the booking flow, requiring separate steps or manual handling, which complicates the process.

From an administrative perspective, many systems lack a centralized dashboard to monitor bookings, analyze user activity, and track revenue. This makes it difficult to manage data efficiently and limits the ability to make informed decisions based on system usage.

User experience is another area where current solutions fall short. Many applications are not designed with intuitive interfaces or smooth workflows, making it difficult for users to navigate between booking, payment, and additional features. The absence of interactive components like visual tracking reduces overall engagement.

Therefore, the problem is not just limited to booking flights but extends to the lack of integration, interactivity, data management, and user-friendly design within a single platform. There is a clear need for a system that combines all these features into one cohesive solution.

This project aims to address these issues by developing a comprehensive Flight Booking and Management System that includes flight selection, seat booking, baggage management, live tracking visualization, and payment integration within a unified interface.

The system focuses on solving the following problems:

1. Lack of integration between booking, seat selection, and baggage management.
2. Absence of real-time or visual flight tracking features.
3. No proper seat locking mechanism leading to booking conflicts.
4. Limited administrative control and lack of analytics dashboard.
5. Poor user experience due to non-interactive and complex interfaces

By addressing these challenges, the project provides a complete, efficient, and user-friendly solution that enhances the overall flight booking experience while ensuring better control, visualization, and system reliability

1.3 Scope of the Project

The scope of this project, focuses on creating a comprehensive, web-based platform that integrates essential flight-related services into a single, user-friendly interface. The system is designed to simplify and enhance the process of booking and managing air travel by combining multiple functionalities such as flight selection, seat booking, baggage management, live tracking visualization, and payment processing.

The technical scope of the project involves developing a responsive and interactive web application using modern front-end technologies such as HTML, CSS, JavaScript, and frameworks like React. The system ensures smooth performance across different devices including desktops, tablets, and smartphones. The application structure is modular, allowing easy maintenance and future enhancements without affecting the overall system

Functionally, the platform enables users to search and select flights based on source, destination, date, and price. It provides an interactive seat selection system where users can choose available seats, along with a seat locking mechanism to prevent conflicts during booking.

A key feature within the scope is the implementation of live flight tracking visualization. Instead of a static display, the system provides an animated representation of the flight journey, improving

user engagement and offering a modern interface experience. This enhances the overall usability and makes the application more interactive.

The project also includes a secure and structured payment module, where users can complete their booking process. All selected options such as seats and food are integrated into the final billing system to ensure accurate cost calculation.

From an administrative perspective, the scope includes a basic analytics dashboard that allows monitoring of bookings, user activity, and revenue. This helps in maintaining centralized control over the system and supports better decision-making.

The user experience scope focuses on providing a clean, intuitive, and easy-to-navigate interface. The system is designed to minimize complexity, ensuring that users can complete the booking process efficiently without confusion. Interactive elements such as progress indicators and visual feedback improve usability and engagement.

In terms of scalability, the system is designed to support future enhancements such as real-time APIs, advanced analytics, or additional travel-related features. However, the current scope is limited to flight booking and management functionalities only.

The limitations of the project include the absence of real-time airline API integration in the current version, meaning that flight data may be static or simulated. Additionally, the live tracking feature is a visual simulation rather than actual GPS-based tracking. The system also focuses only on web-based implementation and does not include a mobile application version at this stage.

To summarize, the scope of the project includes:

1. Development of a web-based flight booking and management system.
2. Integration of seat selection with seat locking functionality.
3. Implementation of baggage selection with dynamic pricing.
4. Inclusion of animated live flight tracking visualization.
5. Secure payment and booking confirmation process.
6. Basic administrative dashboard for monitoring and analytics.
7. User-friendly interface with responsive design.

In conclusion, the project scope defines a focused and practical solution aimed at improving the flight booking experience through integration, interactivity, and simplicity. It provides a strong foundation for future expansion while delivering all essential features required for an efficient and modern flight management system.

1.4 Objectives of the Project

The primary objective of this project is to design and develop an integrated travel management platform that provides users with a seamless and efficient way to plan, manage, and book their travel activities in a single system. The platform combines essential travel services such as flight booking, seat selection, baggage management, and live journey tracking into a unified and user-friendly web application.

A key objective of the system is to simplify the flight booking process by allowing users to search flights, view detailed information, select seats interactively, and choose baggage options based on their requirements. The system ensures that all these functionalities are interconnected, providing a smooth and consistent user experience throughout the booking process.

Another important objective is to enhance user experience through real-time features such as live flight tracking using map integration. This feature visually represents the journey from source to destination with animated flight movement, making the application more interactive and engaging for users. The project also focuses on developing a responsive and intuitive user interface using modern web technologies. The interface is designed to be simple, visually appealing, and easy to navigate for all types of users.

Additionally, the system aims to manage user and booking data efficiently by maintaining structured records of travel details. This helps in organizing information such as passenger data, selected seats, and travel history within a centralized system.

Another objective is to build the application with a modular structure, where each component—such as flight selection, seat booking, and live GPS —functions independently while remaining fully integrated. This approach improves scalability and allows future enhancements to be added easily.

The project also emphasizes automation of the booking workflow, reducing manual effort and minimizing errors. By streamlining processes such as seat allocation, the system improves overall efficiency and user convenience.

Summary of Objectives

1. To develop a unified travel platform for flight booking and management.
2. To implement interactive seat selection and baggage selection systems.
3. To provide live flight tracking using map-based visualization.
4. To design a responsive and user-friendly interface.
5. To manage travel and user data efficiently in a structured system.
6. To ensure modular design for scalability and future enhancements.

In conclusion, the project aims to create a smart, efficient, and interactive travel booking system that enhances user experience while simplifying the overall travel planning process.

1.5 Importance of the Application

The importance of this project lies in its ability to simplify and enhance the flight travel experience through a single, integrated digital platform. Unlike traditional systems where users must rely on multiple steps and disconnected processes, this application brings together all essential components of flight booking—such as flight selection, seat reservation, baggage selection, and live journey tracking—into one unified and user-friendly system.

One of the key contributions of this application is the simplification of the booking workflow. Users can easily search for flights, select their preferred seats visually, and choose baggage options based on their needs within a single interface. This reduces user effort, saves time, and minimizes confusion, making the entire travel planning process smooth and efficient.

Another major aspect of the project's importance is its focus on user experience. The application is designed with an intuitive and interactive interface that allows users to navigate effortlessly, even without technical knowledge. Features like structured layouts, clear options, and step-by-step booking enhance usability and improve overall satisfaction.

A unique feature of this system is the inclusion of live flight tracking using map visualization. This functionality provides users with a real-time view of the journey from source to destination, including animated airplane movement. It adds an interactive and modern touch to the application, making it more engaging compared to traditional booking platforms. Similarly, the seat selection feature enables users to choose their preferred seating arrangement, enhancing comfort and personalization.

From a technical perspective, this project demonstrates the practical implementation of modern web development concepts such as component-based architecture, responsive design, and dynamic data handling. The system is designed to work across multiple devices, ensuring accessibility and consistency for all users.

The application also contributes to better data management by maintaining organized records of user details, booking information, seat allocation, and baggage preferences. This centralized approach improves efficiency and makes it easier to manage travel-related data.

In addition, the modular structure of the system ensures scalability and flexibility. Each module—flight booking, seat selection, and live tracking—works independently while remaining fully integrated. This allows future enhancements, such as adding hotel booking or additional travel services, without affecting the existing system.

From an academic standpoint, the project is significant as it applies software engineering principles to solve a real-world problem. It showcases how technology can be used to automate complex processes and improve user interaction in the travel domain.

Key Points of Importance

1. Simplifies flight booking by integrating all steps into one platform.
2. Enhances user experience through an intuitive and interactive interface.
3. Provides live flight tracking with real-time map visualization.
4. Enables personalized travel through seat and baggage selection systems.
5. Improves efficiency through automation and structured data management.
6. Ensures accessibility with responsive design across devices.
7. Supports scalability with a modular system architecture.
8. Demonstrates real-world application of modern web development techniques.

In conclusion, the importance of this application lies in transforming a traditionally complex flight booking process into a simple, interactive, and efficient digital experience. By combining automation, personalization, and real-time features, the project delivers a smart solution that improves both usability and functionality in modern travel systems.

1.6 Target Users / Audience

The target audience of this project primarily includes users who are looking for a simple, efficient, and interactive flight booking experience. The system is designed to serve individuals who want to manage their travel in a structured and user-friendly way without dealing with complex or time-consuming booking processes.

The primary users are individual travelers such as students, working professionals, and frequent flyers who need a quick and reliable way to book flights. These users benefit from features like easy flight selection, interactive seat booking, customizable baggage options, and live flight tracking. The platform helps them complete all necessary steps within a single application, saving both time and effort.

Another important group of users includes families and group travelers. These users often require coordinated seat selection and baggage planning. The system allows them to choose seats together and manage baggage efficiently, making group travel more organized and convenient. Business travelers also form a significant part of the target audience. They require fast booking, clear information, and minimal complexity. The platform supports their needs by providing a streamlined booking process and quick access to travel details, helping them manage tight schedules effectively.

In addition, the application is suitable for first-time users who may not be familiar with online booking systems. The simple interface, step-by-step flow, and visual features like seat layout and live map tracking make it easy for beginners to understand and use the system without confusion. From a technical and academic perspective, the system can also be used by students, developers, and researchers who are interested in understanding real-world implementation of web-based travel systems. It serves as a practical example of how modern technologies can be applied to solve everyday problems in the travel domain.

Key Target Users

1. Individual travelers (students, professionals, frequent flyers)
2. Families and group travelers requiring coordinated booking
3. Business travelers needing quick and efficient scheduling
4. First-time users seeking a simple and guided booking system
5. Students and developers studying real-world web applications

Conclusion

Overall, the target audience of this project includes a wide range of users who value simplicity, speed, and convenience in flight booking. By focusing on user-friendly design and essential travel features, the system ensures accessibility and usability for both experienced and new users.

CHAPTER -2 SYSTEM REQUIREMENTS

2.1 Hardware Requirements

Hardware requirements form the foundation for the development and execution of this Flight Booking and Management System. They define the physical components required to efficiently design, develop, test, and deploy the application. Since the project is a web-based system with modules such as Flight Selection, Seat Booking, Baggage Selection, and Live Map Tracking, the hardware must support smooth performance, multitasking, and real-time interaction.

During the development phase, a moderately high-performance system is required because developers need to run multiple tools simultaneously, such as code editors, web browsers, and local development servers. The recommended processor is an Intel Core i3/i5 or AMD equivalent with a minimum clock speed of 2.0 GHz to ensure efficient execution of development tasks.

A minimum of 8 GB RAM is required, while 16 GB RAM is recommended for better performance when handling multiple browser tabs, development environments, and real-time UI updates. Since the project includes interactive features like seat selection layouts and live map visualization, sufficient memory ensures smooth rendering and execution.

For storage, a minimum of 256 GB SSD is recommended. Solid-State Drives (SSD) provide faster data access compared to traditional hard drives, improving application loading time and overall system responsiveness during development and testing.

A stable internet connection with a minimum speed of 5–10 Mbps is required for downloading dependencies, testing APIs (if any), and ensuring proper functioning of live map tracking features. Internet connectivity is also essential for deployment and version control operations. The system should support operating systems such as Windows 10/11, Ubuntu Linux, or macOS, all of which are compatible with modern web technologies. A Full HD monitor (1920×1080 resolution or higher) is recommended for better visualization of user interface components, especially for seat layout and map features.

From the end-user perspective, the hardware requirements are minimal since the system is browser-based. Users can access the application using any device such as a desktop, laptop, tablet, or smartphone with a basic configuration of a dual-core processor, 4 GB RAM, and a modern web browser. The system is designed to be lightweight and responsive, ensuring smooth performance even on low-end devices.

In conclusion, the hardware requirements for this project ensure efficient development, reliable deployment, and smooth user experience. The system is optimized to run on both developer machines and standard user devices, making it accessible, scalable, and suitable for real-world applications.

2.2 Software Requirements

The software requirements define the digital environment required to develop, run, and maintain the Corporate Travel Management Portal efficiently. Since the system is a web-based application, it relies on modern technologies to ensure smooth operation, real-time updates, and secure handling of travel data.

The operating system plays a key role in development and deployment. The application can be developed using Windows 10/11, macOS, or Linux. For deployment, Linux-based systems are preferred due to their stability, security, and cost-effectiveness. Developers may use Windows or macOS for ease of coding and testing.

A web server is necessary to host the application and handle client requests. Tools like Apache HTTP Server or Nginx can be used in production environments. For local development, XAMPP or WAMP servers can simulate real server behavior and help in testing the application before deployment.

The database management system is essential for storing and managing all application data, including employee details, travel bookings, expenses, and reports. MySQL is a suitable choice for this project as it is reliable, easy to use, and supports structured data management effectively.

The frontend of the application is developed using HTML, CSS, and JavaScript. These technologies create a user-friendly interface for employees and administrators. Responsive design techniques ensure that the portal works smoothly across devices such as desktops, tablets, and mobile phones.

For development, Visual Studio Code (VS Code) is used as the primary code editor. It provides useful features like syntax highlighting, debugging tools, and extension support. Git and GitHub are used for version control to manage code changes and enable collaboration.

Testing tools are used to ensure the system functions correctly. Basic testing can be done using browser developer tools, while tools like Postman can be used to test backend APIs and ensure proper data communication.

Security is an important part of the software requirements. The system ensures secure login using authentication mechanisms and encrypts sensitive data such as passwords.

Basic validation techniques are used to prevent unauthorized access and protect user data.

The application follows a modular structure, where different components such as travel booking, expense tracking, and reporting are managed separately but connected through a central system. This improves maintainability and allows future enhancements.

In conclusion, the software requirements of the Corporate Travel Management Portal focus on simplicity, efficiency, and security. By using standard web technologies and structured design, the system provides a reliable platform for managing corporate travel activities effectively.

2.3 Development Tools and Frameworks

The development of the Corporate Travel Management Portal requires a structured set of tools and frameworks to ensure efficient coding, testing, and deployment. These tools help in building a reliable, user-friendly, and maintainable system that supports employee travel requests, expense tracking, and administrative monitoring.

At the front-end level, the user interface is designed using HTML, CSS, and JavaScript. HTML provides the basic structure for web pages such as login forms, travel request forms, dashboards, and reports. CSS is used to enhance the visual appearance of the application, making it clean and professional. Features like Flexbox, Grid, and media queries are used to ensure the system is responsive across different devices.

JavaScript is used to add interactivity to the system. It handles form validation, dynamic updates of travel data, and improves user experience without requiring page reloads. Basic JavaScript or modern ES6 features can be used to make the interface more responsive and interactive.

Visual Studio Code (VS Code) is used as the primary development environment. It supports multiple programming languages and provides useful features such as debugging tools, extensions, and Git integration, making development easier and more organized.

Version control is handled using Git and GitHub. Git helps in tracking code changes and maintaining different versions of the project, while GitHub allows developers to store code online and collaborate efficiently. It also helps in managing updates and preventing data loss.

The system includes dashboards for monitoring travel expenses and employee activities. Simple chart libraries may be used to display graphical reports such as monthly expenses, travel frequency, and budget analysis. These visualizations help management make informed decisions.

In conclusion, the development tools and frameworks used in the Corporate Travel Management Portal are selected to ensure simplicity, efficiency, and reliability. These tools support the complete development lifecycle, from coding and testing to deployment, resulting in a well-structured and user-friendly system.

CHAPTER-3 TECHNOLOGY STACK

3.1 Front-End Design

The front-end of the NextGen Travel Intelligence and Automation Portal represents the user interface through which employees and administrators interact with the system to manage corporate travel activities. It is a crucial layer of the application as it directly affects usability, user experience, and efficiency. The front-end is designed to be simple, responsive, and user-friendly so that users can easily submit travel requests, track approvals, and manage expenses without confusion.

The front-end is developed using HTML5, CSS, JavaScript (ES6+), and optionally React.js. These technologies work together to create a dynamic and interactive web application that performs efficiently across various devices including desktops, tablets, and mobile phones.

HTML5 is used to structure the web pages of the application. It defines key components such as login forms, travel request forms, approval sections, expense tracking tables, dashboards, and report views. Semantic HTML elements improve readability and ensure proper organization of content across the system.

CSS is used for styling and layout design. It enhances the visual appearance by applying suitable colors, fonts, and spacing. Features such as Flexbox, Grid, and media queries ensure that the interface is fully responsive and adapts smoothly to different screen sizes. The design follows a clean and professional theme suitable for corporate environments.

JavaScript (ES6+) is used to add interactivity to the application. It enables features such as form validation, dynamic updates of travel requests, real-time status tracking, and interactive dashboards. JavaScript ensures a smooth user experience by reducing unnecessary page reloads. React.js, if implemented, enhances the scalability and maintainability of the front-end. It follows a component-based architecture, where the interface is divided into reusable components such as Login Page, Dashboard, Travel Request Form, Approval Panel, Expense Tracker, and Reports Module. React efficiently updates only required parts of the interface using the Virtual DOM, improving performance.

Navigation between different sections such as Dashboard, Travel Requests, Expense Management, and Reports is handled efficiently (using React Router if React is used), providing a seamless Single Page Application (SPA) experience without full page reloads.

State management is handled using simple JavaScript or React's Context API (if React is used). It helps manage user data such as login sessions, travel details, and expense records across

different components.

In conclusion, the front-end design of the Corporate Travel Management Portal provides a clean, responsive, and efficient interface. The use of modern web technologies ensures that the system is user-friendly, scalable, and suitable for managing corporate travel operations effectively.

3.2 Database Design

The database design of the NextGen Travel Intelligence and Automation Portal plays a vital role in managing all the data related to employees, travel requests, approvals, expenses, and reports. It acts as the backbone of the system by ensuring that all information is stored in a structured, secure, and efficient manner.

The system uses MySQL as the database management system because it is reliable, easy to use, and supports relational data effectively. It helps maintain data integrity through the use of primary keys and foreign keys, which establish relationships between different tables in the system.

The database consists of several important tables. The Employee table stores details such as employee ID, name, email, password, department, and contact information. This table is used for user authentication and identification within the system.

The Approval table manages the approval process. It stores details such as approval ID, request ID, approver information, approval status, and remarks. This ensures that every travel request goes through a proper approval workflow.

The Expense table stores expense-related information such as expense ID, request ID, travel cost, accommodation cost, food expenses, and total amount. This helps in tracking and managing travel budgets effectively.

Relationships between tables are clearly defined:

- One employee can create multiple travel requests.

- Each travel request is linked to one approval record.

- Each travel request is linked to one expense record

In conclusion, the database design is kept simple and directly aligned with the system's core features—travel request management, approval handling, and expense tracking—ensuring efficient performance and easy maintenance.

3.3 Version Control

Version control plays a crucial role in the development of the Airplane Booking System, as it helps in managing code efficiently, tracking changes, and maintaining different versions of the project throughout its lifecycle. It ensures smooth collaboration among developers and maintains the stability and reliability of the system.

The project uses Git as the version control system and GitHub as the remote repository for storing and managing the project code. Git allows developers to keep track of every modification made to the codebase, while GitHub provides a secure cloud platform for collaboration and backup.

The development process is organized using multiple branches. Each module of the Airplane Booking System is developed in separate branches, such as User Authentication, Flight Search, Seat Selection, Booking Management, Payment Processing, and Ticket Generation. This approach prevents conflicts and ensures that changes in one module do not affect other parts of the system.

Version control also acts as a backup system. In case of any errors or failures, previous versions of the project can be restored ensuring that development is not disrupted.

In conclusion, Git and GitHub help in maintaining proper organization, collaboration, security, and version tracking in the Airplane Booking System, making the development process efficient and reliable.

3.4 APIs or External Services

APIs (Application Programming Interfaces) and external services play a key role in the Airplane Booking System, as they enable communication between different modules and support real-time data processing. They help the system provide smooth interaction for users while performing operations such as searching flights, booking tickets, and making payments.

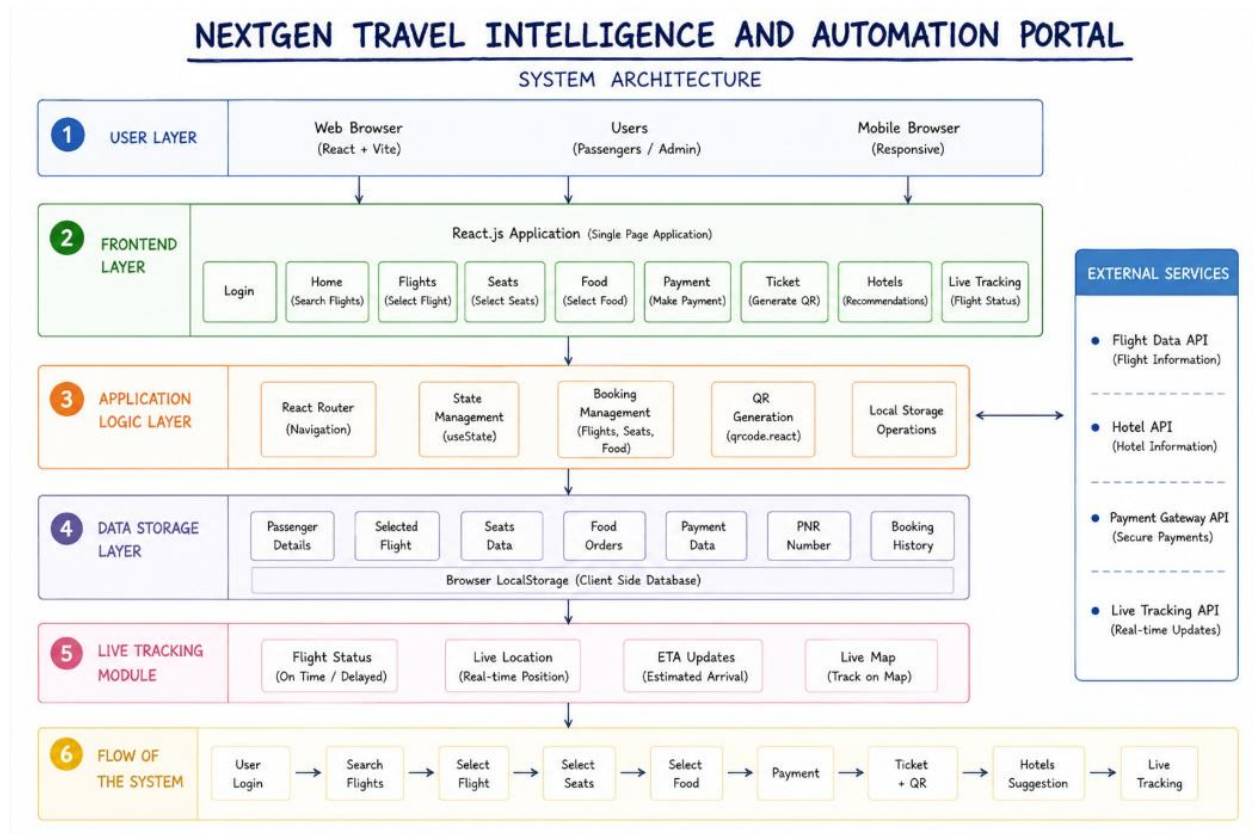
The system includes internal APIs for the following core functionalities:

- User Authentication
- Flight search and availability checking
- Booking management (ticket booking and cancellation)
- Passenger details handling
- Payment processing

To ensure performance and reliability, APIs are tested using tools like Postman. Secure communication is maintained using HTTPS protocol, and sensitive information such as API keys and credentials are stored using environment variables. In conclusion, APIs and external services improve the overall efficiency of the Airplane Booking System by enabling real-time communication, secure transactions, and smooth integration between different components of the system

CHAPTER -4 SYSTEM ARCHITECTURE

High-level architecture diagram



4.1 Description of Each Layer (Frontend, Database)

The NextGen Travel Intelligence and Automation Portal is designed using a layered architecture approach that integrates multiple travel-related services into a single unified system. The architecture is structured to provide an efficient, scalable, and user-friendly environment where passengers can perform various travel activities such as flight booking, seat reservation, food ordering, payment processing, hotel recommendations, ticket generation, and live flight tracking. By combining these services within one platform, the system eliminates the need for users to switch between multiple applications and provides a seamless travel experience.

The system begins with the user interaction layer, where passengers access the application through web browsers or mobile devices. The user interface is developed using React.js and Vite, ensuring a responsive and dynamic experience across different screen sizes. Through this interface, users can create accounts, log in securely, search for flights, compare available options, select preferred

seats, customize food choices, and complete their bookings. The frontend is designed to provide smooth navigation and real-time updates, allowing users to interact with the system efficiently.

The frontend layer serves as the primary communication channel between users and the application. It contains several interconnected modules including Login, Home, Flight Search, Seat Selection, Food Ordering, Payment Processing, Ticket Generation, Hotel Recommendations, and Live Tracking. Each module is designed as a reusable React component, which improves maintainability and scalability. React Router is utilized to manage page navigation, while state management techniques are used to preserve user selections throughout the booking process. This architecture ensures that user data flows seamlessly between different pages without requiring constant page reloads.

The application logic layer is responsible for handling the core operations of the system. This layer manages booking workflows, validates user inputs, calculates ticket costs, processes food selections, generates PNR numbers, and creates QR-code-based tickets. It also controls the overall sequence of actions performed by users during the booking process. Various calculations such as total fare, food costs, seat charges, and booking summaries are processed within this layer before being presented to the user. By separating business logic from the user interface, the system achieves better organization and maintainability.

To enhance the functionality of the platform, the architecture integrates multiple external services. Flight APIs provide route information, airline details, schedules, and travel-related data. Hotel APIs are used to display accommodation recommendations near the destination city. Payment Gateway APIs ensure secure transaction processing during booking confirmation. Live Tracking APIs provide real-time flight status updates, allowing passengers to monitor their journey after ticket confirmation. The integration of these external services enables the platform to deliver accurate, reliable, and real-time information to users.

The system utilizes Browser Local Storage as its primary client-side storage mechanism. Important booking information such as passenger details, selected flights, seat reservations, food orders, payment records, booking history, and generated PNR numbers are stored locally within the browser. This approach ensures that data remains available even when the page is refreshed and allows users to access booking information without requiring a dedicated backend database. The stored data is also utilized for generating tickets, maintaining booking history, and displaying analytical information within the system.

One of the key features of the architecture is the QR-code-based ticket generation mechanism. After successful payment confirmation, the system automatically generates a digital ticket containing passenger details, travel information, booking reference number, and QR code. This

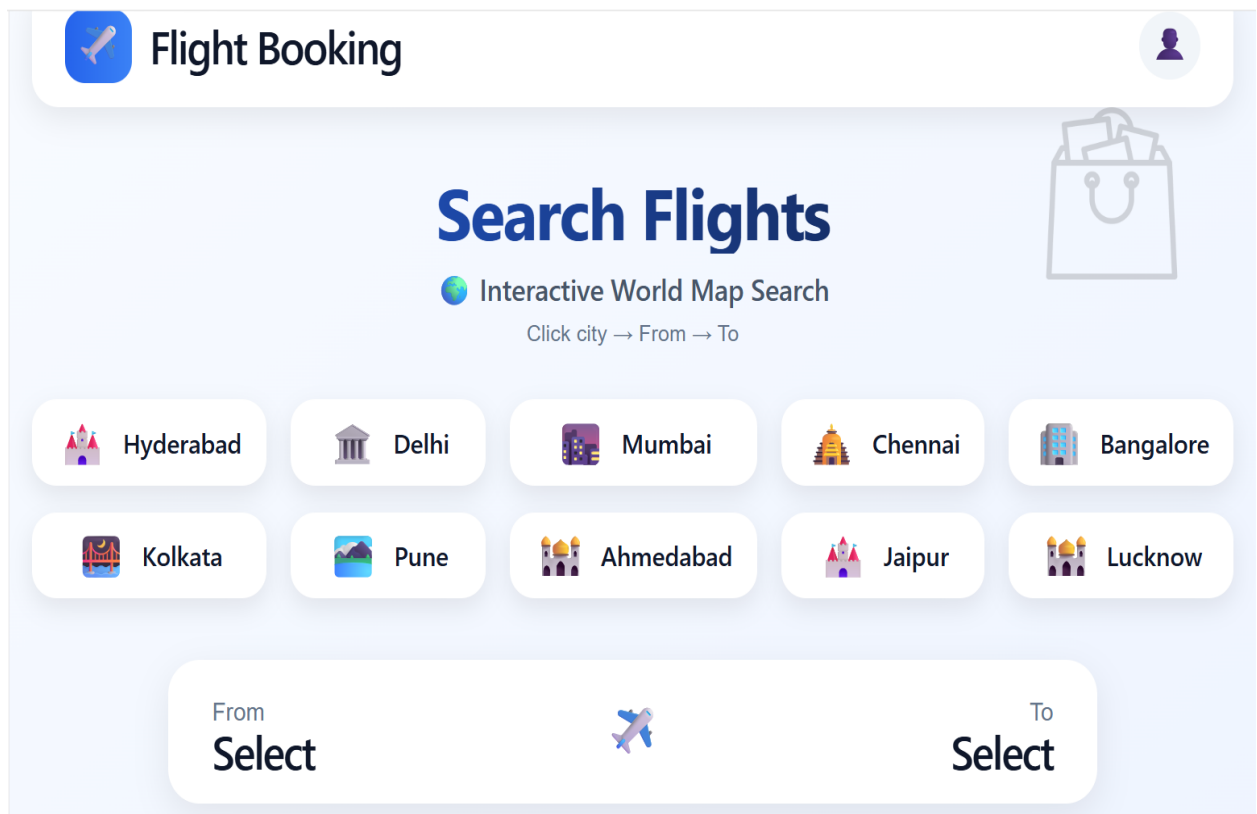
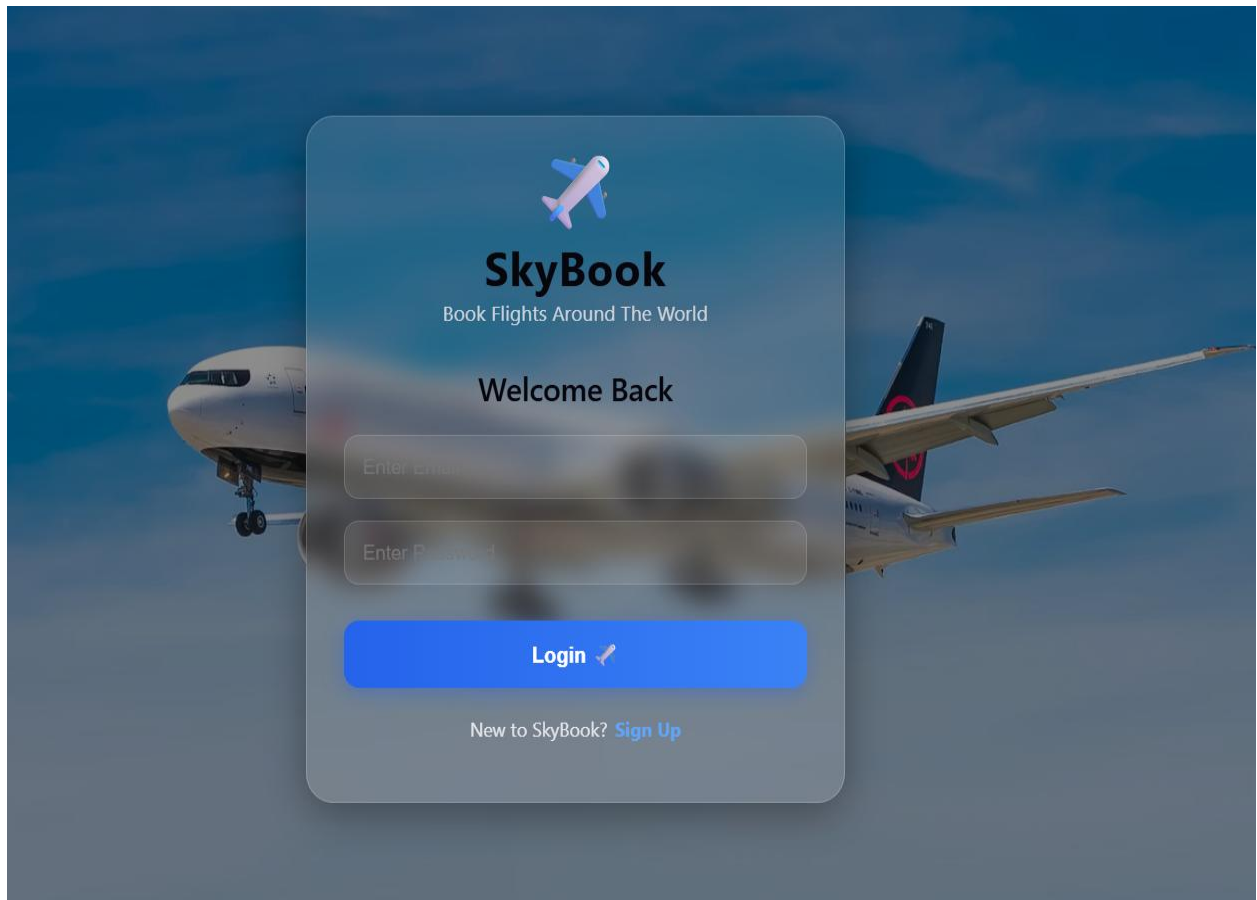
QR code can be scanned for quick verification and serves as a modern digital alternative to traditional paper tickets. The ticket generation process enhances convenience while improving the overall user experience.

The hotel recommendation module further extends the functionality of the platform by providing accommodation options near the user's destination. After booking a flight, passengers can explore nearby hotels, view prices, ratings, and locations, and make accommodation decisions directly within the platform. This feature transforms the application from a simple flight booking system into a more comprehensive tourism management solution.

Another important component of the architecture is the live tracking module. This module allows passengers to monitor the current status of their flights after ticket confirmation. Real-time information such as flight location, estimated arrival time, route details, and flight status updates are displayed through an interactive tracking interface. By providing continuous travel updates, the live tracking system improves transparency and enhances passenger confidence throughout the journey.

In conclusion, The NextGen Travel Intelligence and Automation Portal combines modern frontend technologies, efficient application logic, client-side storage mechanisms, and third-party service integrations to provide a complete travel and tourism solution. The architecture is designed to support scalability, maintainability, usability, and real-time functionality, making it suitable for modern travel booking applications. Through its integrated approach, the platform successfully delivers flight booking, accommodation assistance, digital ticketing, and live tracking services within a single unified ecosystem.

CHAPTER -5 APPENDICES



Available Flights

Mumbai → Chennai

All Timings ▾



IndiGo
Boeing 737

Mumbai → Chennai

Morning (10:00 AM)
Sat, 6 Jun

₹4353

Select



Air India
Boeing 737

Mumbai → Chennai

Morning (9:00 AM)
Sat, 6 Jun

₹3646

Select



Air India
Boeing 737

Mumbai → Chennai

Afternoon (3:30 PM)
Sat, 6 Jun

₹4287

Select



Vistara
Boeing 737

Mumbai → Chennai

Afternoon (4:30 PM)
Sat, 6 Jun

₹4272

Select



SpiceJet
Boeing 737

Mumbai → Chennai

Evening (6:30 PM)
Sat, 6 Jun

₹5182

Select



SpiceJet

Mumbai → Chennai

Evening (8:00 PM)

₹4667



Seat Selection

Choose your favorite seat

Mumbai → Chennai

Air India • AI-746 • Sat, 6 Jun

Available Selected Occupied Premium

A1	B1	C1	D1	E1	F1	G1	H1
A2	B2	C2	D2	E2	F2	G2	H2
A3	B3	C3	D3	E3	F3	G3	H3
A4	B4	C4	D4	E4	F4	G4	H4
A5	B5	C5	D5	E5	F5	G5	H5
A6	B6	C6	D6	E6	F6	G6	H6
A7	B7	C7	D7	E7	F7	G7	H7
A8	B8	C8	D8	E8	F8	G8	H8
A9	B9	C9	D9	E9	F9	G9	H9
A10	B10	C10	D10	E10	F10	G10	H10
A11	B11	C11	D11	E11	F11	G11	H11
A12	B12	C12	D12	E12	F12	G12	H12
A13	B13	C13	D13	E13	F13	G13	H13
A14	B14	C14	D14	E14	F14	G14	H14
A15	B15	C15	D15	E15	F15	G15	H15

Booking Summary



Air India

AI-746 • Boeing 737

Flight Price ₹4287

Seat Charges ₹0

Seat(s) 0

Total Amount

₹4287

Passenger Details

Select seats to add passengers

Selected Seats

No seats selected

[Continue to Food →](#)

In-Flight Food Selection

Page 26



SkyBook Boarding Pass

Your booking has been confirmed





Payment Successful

Your ticket is confirmed and ready for boarding.

Mum
Mumbai


Air India
6E 123

Che
Chennai



Journey Date
31 May 2026
Tuesday



Departure Time
3:30 PM
On Time



Selected Seats
E6
Economy



PNR Number
SKY898552
Booking Reference



Food Charges
₹450
Added

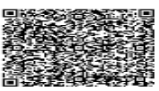


Total Paid
₹5537
Paid via Card



Boarding Information

Please arrive 2 hours before departure.
Carry valid ID proof for verification.



 [Print Ticket](#)

 [Back To Home](#)

Airport Details (Detailed Structure)

Airport Map



 Duty Free

 Security Check

 Check-in Counter

 Entrance

 Washrooms

 Food Court

 Lounge

 Charging Station

 Duty Free Shop

 Medical Help

 Washrooms

 Food Court

 Lounge

 Charging Station

 Medical Help

 Your Gate
T1-G5



Route Details

- Entrance
- Check-in Counter
- Security Check
- Duty Free
- Terminal
- **T1-G5**

Air India

Passenger: madhu
PNR: SKY263740
From: Mumbai
To: Ahmedabad
Departure: 01:30 PM
Arrival: 03:45 PM
Boarding Time: 12:45 PM

 [View Nearby Hotels](#)



Nearby Hotels

Best hotels near your destination



Taj Palace Hotel

★ 4.8

📍 Near Airport

Distance
🚗 2.5 km

Reach Time
🕒 8 mins

₹4500

/night

Book Now



Grand Luxury Stay

★ 5

📍 City Center

Distance
🚗 5 km

Reach Time
🕒 15 mins

₹3800

/night

Book Now



Royal Inn

★ 4.9

📍 Heer's Cottage

Distance
🚗 7 km

Reach Time
🕒 20 mins

₹3200

Book Now



Airport Residency

★ 4.7

📍 Airport Road

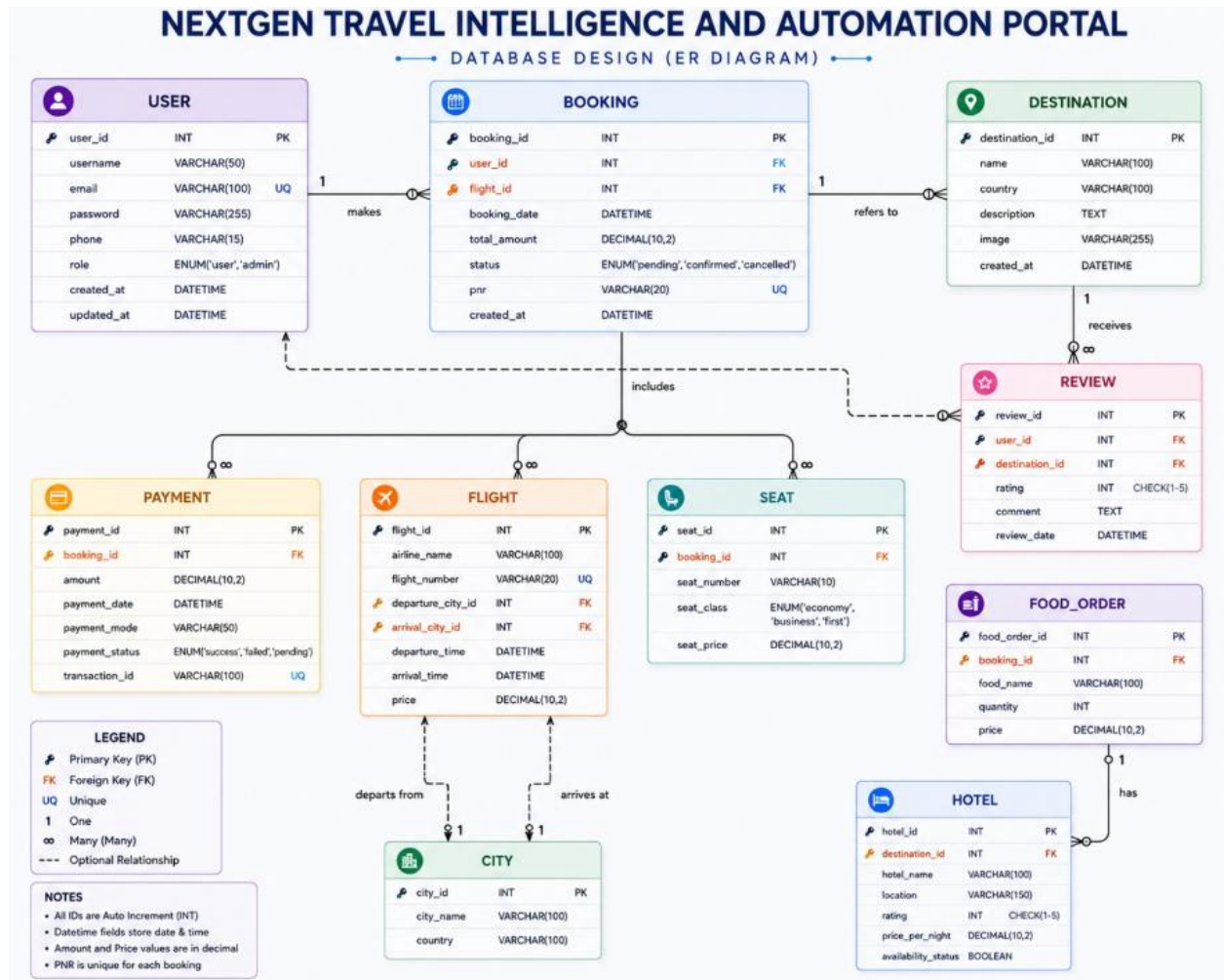
Distance
🚗 1.8 km

Reach Time
🕒 5 mins

₹2900

Book Now

CHAPTER -6 DESIGN



CHAPTER -7 IMPLEMENTATION

Section 6.1 – Module-Wise Implementation

1. Authentication and User Management Module

This module handles user login and basic session entry into the system. It ensures that only authorized users can access the travel booking platform. User details such as credentials are validated before granting access to the Home Module. This acts as the entry point of the application.

2. Home and Travel Selection Module

The Home Module serves as the central dashboard where users initiate their travel planning. Users can search and select travel routes by entering source and destination details. It acts as the navigation hub that connects users to the flight listing module and stores initial booking context.

3. Flight Search and Listing Module

This module displays available flights based on user-selected routes and travel details. Each flight includes information such as airline name, flight number, aircraft type, price, time, and images. Users can compare multiple flights and select a preferred option. The selected flight data is passed to the next module using route state management.

4. Seat Selection Module

The Seat Selection Module is one of the core components of the system. It provides an interactive aircraft seating layout divided into First Class, Business, and Economy sections. Key functionalities include:

- Dynamic seat generation (A–H rows 1–15)
- Randomly generated occupied seats
- Premium seat classification
- Seat selection and deselection
- Real-time price calculation based on selected seats
- Passenger detail input (name and age per seat)
- Storage of booking data using localStorage (PNR, passenger details, selected seats, seat charges)

This module ensures a realistic airline booking experience similar to real-world systems.

5. Passenger Information Module

This module automatically generates passenger input forms based on selected seats. Each selected seat is mapped to a passenger entry form where users must enter name and age. The system validates that all passenger details are completed before proceeding to the next step.

6. Food Selection Module

After seat selection, users are redirected to the in-flight food selection module. This module allows users to choose meals additional food items for their journey. The selected food items contribute to the total booking cost and are carried forward to the payment ticket generation process.

7. Payment and Billing Module

This module calculates the final fare by combining:

- Base flight price
- Seat selection charges
- Food charges

It ensures transparent billing before confirmation. Once payment is completed, booking details are finalized and stored in the system for ticket generation.

8. Ticket Generation Module

The Ticket Module generates a digital boarding pass after successful booking. It includes:

- Passenger details
- Flight route (From → To)
- Airline information
- Seat numbers
- PNR generation (SKY + random number)
- Total fare summary
- QR Code generation for digital verification
- Boarding instructions and Boarding Pass

This module also retrieves data from localStorage to ensure persistence across pages.

9. Hotel Module

The Hotel Module allows users to search and book hotel rooms as part of the travel system. After completing flight booking, users can move to this module to select accommodation based on their travel needs.

6.2 – Front-End Logic (UI Rendering and State Management)

The front-end of the NextGen Travel Intelligence and Automation Portal provides a smooth and interactive interface for managing flight search, seat selection, food ordering, payment, and ticket generation. It focuses on delivering a seamless user experience where all travel-related actions are handled in a single integrated workflow.

Built using a React component-based architecture, the system is divided into reusable modules such as login, flight listing, seat selection, food selection, payment, and ticket pages. This modular design ensures better code organization, maintainability, and scalability of the application.

State management is handled using React `useState`, `useEffect`, and React Router (`useLocation` and `navigate`). Booking details such as flight information, selected seats, passenger details, food selections, and total fare are dynamically updated and passed between components. Additionally, `localStorage` is used to persist important data like PNR, passenger name, selected seats, and pricing, ensuring data availability across different pages.

The seat selection module provides real-time UI updates for available, occupied, selected, and premium seats, while automatically calculating seat-based pricing. The passenger form dynamically generates input fields based on selected seats, ensuring accurate data entry.

The payment and ticket modules display dynamically calculated totals and generate a digital boarding pass with a QR code for verification. This enhances the system by enabling a paperless and automated travel experience.

Overall, the front-end ensures a responsive, modular, and user-friendly interface, making the travel booking process efficient, interactive, and easy to manage.

6.3 API Integration

The API Integration layer of the NextGen Travel Intelligence and Automation Portal acts as the communication bridge between the front-end React application and the back-end services or data handling logic. It ensures smooth data flow across different modules such as flight selection, seat booking, passenger management, food ordering, payment processing, and ticket generation.

In the current implementation, data is passed between pages using React Router (`useLocation` and `navigate`). This allows the application to share booking details like flight information, selected

seats, and passenger data without page reloads.

The system also uses `localStorage` to store important information such as PNR number, passenger details, seat selection, and pricing. This helps maintain data even after refreshing the page.

Basic validation is handled within the front-end to ensure correct data flow between modules, such as checking passenger details and seat selection before moving to the next step.

Overall, the system is designed in a modular and API-ready structure, making it easy to integrate full backend services in future upgrades.

6.1 Authentication and Authorisation

The Authentication and Authorisation module of the NextGen Travel Intelligence and Automation Portal ensures secure access to the application and protects user booking data. It manages user login and controls access to different parts of the system.

In the current implementation, the system uses a simple login module (Login page) where user credentials are verified before allowing access to the platform. This acts as the entry point to the travel booking system.

After successful login, user session data is maintained using the React state and `localStorage`. This helps in preserving user information across different modules such as flights, seats, food, and ticket pages.

Authorisation is handled at the UI level by controlling navigation between pages. Only logged-in users can move through the booking flow (home → flights → seats → food → payment → ticket), ensuring a guided and restricted access path.

Basic validation is applied in forms such as login and passenger details to ensure correct input before proceeding to the next step. Error handling is managed using alerts and conditional checks within components.

Overall, the system provides a simple and secure front-end based authentication flow, which can be further enhanced with JWT and backend integration in future development.

CHAPTER 8- FEATURES

7.1 List of Main Features

The NextGen Travel Intelligence and Automation Portal is a unified travel management system designed to simplify and automate the entire travel booking process. It brings together all major travel services into a single platform, allowing users to plan, book, and manage their journey in a smooth and efficient manner.

The system mainly includes a Flight Search and Booking feature, where users can explore and select flights based on their travel preferences such as route, date, airline, and price. Once a flight is selected, the Seat Selection feature allows users to choose seats from an interactive layout, with real-time updates for available, occupied, and premium seats

To enhance personalization, the platform includes a Passenger Management feature that collects individual passenger details for each selected seat. Along with this, the In-Flight Food Selection feature allows users to choose meals during the booking process, making the travel experience more customized.

The Payment and Billing feature calculates the total cost including flight fare, seat charges, and food selections, ensuring transparency before final confirmation. After successful booking, the system generates a Digital Ticket with PNR and QR Code, providing a paperless and easily verifiable boarding pass.

The portal also uses localStorage-based data persistence to maintain booking details across pages and ensure smooth navigation through the booking flow. A structured step-by-step navigation system guides users from login to final ticket generation, ensuring an easy and organized workflow.

The application follows a Structured Navigation Workflow, guiding users step-by-step from login to flight selection, seat booking, food selection, payment, and final ticket generation.

Finally, the system is built using a Modular and Scalable Architecture, making it easy to maintain, extend, and integrate with backend APIs in the future for real-time data and advanced functionalities.

7.2 How Each Feature Works

1. Flight Selection Feature

User Perspective:

The system starts with the flight listing page where users can browse available flights based on origin, destination, airline, date, and price. Once a user selects a flight, the details such as airline name, flight number, aircraft type, and base fare are passed to the next module using React Router navigation state.

Technical Description:

This module uses React state and `useNavigate()` to transfer selected flight data between pages. The selected flight object is stored in `location.state`, ensuring smooth page transitions without page reload. This avoids database re-fetching and improves performance by carrying data through route navigation.

2. Seat Selection Feature:

User Perspective:

After choosing a flight, users are taken to the seat selection screen where they can visually select seats from an aircraft layout. Seats are color-coded to indicate availability, selection status, and premium pricing. Users can pick multiple seats, and each selected seat dynamically generates a passenger form. When a user clicks on an available seat, it becomes highlighted and added to their selection list. Multiple seat selection is supported, allowing group bookings or family reservations. Occupied seats cannot be selected, ensuring realistic booking constraints.

Technical Description:

Seat generation is handled using nested loops for rows (1–15) and columns (A–H). Random occupied seats are generated using `useEffect()` with a shuffled array. Seat selection is managed using React `useState()` arrays. When a seat is selected, state updates trigger re-rendering of UI and passenger form creation dynamically.

3. Passenger Management Feature

User Perspective:

For every selected seat, the system automatically creates a passenger input form where users must enter name and age. Each seat corresponds to one passenger, ensuring accurate ticket mapping. The interface dynamically adjusts when seats are added or removed. If a seat is deselected, the corresponding passenger form is also removed, maintaining synchronization between seat selection and passenger data.

Technical Description:

Passenger data is stored in a React state array. A `useEffect()` hook synchronizes passenger objects with selected seats. On input change, state is updated immutably using array spreading. Validation checks ensure that no empty name or age fields are allowed before proceeding.

4. Pricing and Summary Feature

User Perspective:

The booking summary section provides a real-time breakdown of the total cost. It displays the base flight price, seat selection charges, number of seats selected, and the final payable amount. Users can clearly review all charges before proceeding further.

Technical Description:

Pricing is calculated using derived state variables. Seat cost is computed as `selectedSeats.length * 800`, and total amount is the sum of base flight price and seat charges.

React re-renders automatically update the summary whenever seat selection changes.

5.Booking Continuation and Data Persistence Feature

User Perspective:

After confirming seat selection and passenger details, the user proceeds to the next module (Food Selection). All booking information is preserved and carried forward without data loss.

Technical Description:

Booking data is stored using `localStorage`, including passenger details, seat selection, seat price, and a generated PNR number. This ensures persistence even if the page reloads. The PNR is generated using a random number algorithm prefixed with "SKY".

6. Food Selection Feature

User Perspective:

After seat selection, users can optionally choose in-flight food items. Selected items are added to the final bill before payment. The food module allows users to optionally select in-flight meals. Users can browse available food items and add them to their booking. The selected food items are added to the final bill, giving users flexibility to customize their travel .

Technical Description:

Food selections are managed through React state and passed between pages using `navigate()` with state. The selected food items contribute to the final price calculation and are stored in `localStorage` under `foodTotal`. This ensures continuity in pricing when transitioning to the payment module.

7.Payment Feature

User Perspective:

The payment module displays a complete breakdown of all charges including flight fare, seat selection cost, and food charges. Users can review the final amount before confirming payment.

Technical Description:

The payment process is implemented as a front-end simulation without external gateway integration. Once the user confirms payment, booking details are finalized and stored. The system ensures that all required booking data is available in `localStorage` before generating the ticket.

8. Ticket Generation Feature

User Perspective:

After successful payment, the system generates a digital boarding pass containing all booking details. This includes passenger names, selected seats, PNR number, flight route, and total amount paid. A QR code is also displayed, which can be scanned for quick access to booking information.

Technical Description:

The ticket module retrieves data from `localStorage` and route state. The QR code is generated using the `QRCodeCanvas` component from the `qrcode.react` library. The encoded QR data includes flight details, passenger list, seat numbers, and booking reference (PNR).

9. Navigation and Data Flow Architecture

User Perspective:

The application follows a smooth step-by-step flow:

Flight Selection → Seat Selection → Passenger Details → Food Selection → Payment → Ticket Generation

Users experience a continuous workflow without page reloads.

Technical Description:

The system is built as a Single Page Application (SPA) using React Router. Navigation between modules is handled using `BrowserRouter`, `Routes`, and `Route`. Data is passed using a combination of:

- `location.state` (temporary navigation data)
- `localStorage` (persistent storage)

10. State Management Feature

User Perspective:

All interactions such as seat selection, passenger input, and pricing updates happen instantly without delay. The system feels responsive and interactive.

Technical Description:

State management is handled entirely using React Hooks:

- `useState()` for dynamic data
- `useEffect()` for side effects and synchronization

Changes in seat selection immediately trigger updates in passenger forms and pricing calculations. This ensures real-time synchronization across the application without requiring external state management libraries.

CHAPTER -9 CHALLENGES & LIMITATIONS

10.1 Issues Faced During Development

During the development of the NextGen Travel Intelligence and Automation Portal, one of the main challenges was maintaining smooth data flow between multiple modules such as flight selection, seat booking, food selection, payment, and ticket generation. Since each module depends on the previous one, ensuring proper transfer of data between routes using React Router required careful handling.

Another issue was managing state consistency across the application. With multiple React Hooks like `useState` and `useEffect`, synchronizing seat selection with passenger details sometimes caused mismatches. For example, when a seat was removed, the corresponding passenger data also needed to be updated correctly to avoid errors.

Data persistence was also a challenge because the application does not use a backend database. Important booking details such as seats, passenger names, and PNR were stored in `localStorage`. Initially, data was lost on refresh or navigation, so proper storage and retrieval logic had to be implemented.

The seat selection module created additional complexity due to dynamic rendering of a large seat layout. Managing occupied, selected, and premium seats in real time required efficient condition checks to ensure correct UI updates and prevent selection of unavailable seats.

Another issue was in pricing calculation, where the total amount depends on both flight fare and selected seats. Any delay in state updates could lead to incorrect pricing, so the logic had to be made fully reactive to ensure accuracy.

Finally, minor challenges were faced in maintaining UI consistency and debugging navigation flow issues. Ensuring that users follow the correct booking sequence (flight → seat → food → payment → ticket) was important to prevent missing or undefined data during ticket generation.

10.2 Solutions Applied

During the development of the NextGen Travel Intelligence and Automation Portal, several structured solutions were implemented to overcome challenges related to data flow, state management, and application performance. Since the system consists of multiple interconnected modules (Flights, Seats, Food, Payment, and Ticket), a consistent navigation and data transfer strategy was established using React Router and navigation state to ensure smooth communication between pages.

To solve state management issues, React Hooks such as `useState` and `useEffect` were carefully structured to maintain synchronization between seat selection and passenger details. Whenever a seat was added or removed, the passenger array was dynamically updated to ensure both data sets remained consistent and error-free.

For handling data persistence, `localStorage` was implemented as the primary solution. All critical booking details such as selected seats, passenger information, PNR number, and pricing data were stored locally in the browser. This ensured that even after page refresh or navigation, the booking data remained intact and accessible throughout the user journey.

In the seat selection module, dynamic rendering and conditional logic were optimized to improve performance. Randomly generated occupied seats and premium seat categories were clearly separated using dedicated arrays and styling functions. This helped in improving UI clarity and preventing invalid seat selections.

Pricing calculation issues were resolved by introducing fully reactive computation logic. The total amount was recalculated automatically whenever there was a change in seat selection. This ensured accurate and real-time updates for users without manual intervention.

To maintain a proper booking flow, validation checks were introduced at each stage of the application. Users were required to complete previous steps (flight selection and seat selection) before moving forward, preventing incomplete or inconsistent booking data.

For ticket generation, reliable data retrieval mechanisms were implemented using a combination of `localStorage` and route state. Proper checks were added to handle missing or undefined values, ensuring that the final boarding pass always displayed complete and accurate information.

Finally, UI consistency and usability were improved through structured inline styling and reusable components. Continuous testing across different user flows helped refine the interface and ensure a smooth, responsive, and user-friendly experience across the entire application.

10.3 Current Limitations or Known Bugs

One of the key limitations of the NextGen Travel Intelligence and Automation Portal is its dependency on external APIs for flight, train, and hotel data. Since these services are controlled by third-party providers, occasional issues like downtime, slow responses, or rate limits can affect real-time data accuracy and system performance.

Another limitation is data synchronisation across modules. Because multiple services are integrated, there may be slight delays in updating booking status, availability, or pricing. This can sometimes lead to temporary mismatches between displayed information and actual backend data.

From a security and authentication perspective, JWT-based login is effective but may occasionally lead to unexpected session expiry issues. In long inactive sessions, users may be logged out automatically, which affects user experience but does not compromise security.

The payment integration module is reliable but has limited flexibility in currency support and depends heavily on external payment gateways. Any restrictions or failures from these gateways can temporarily delay transactions or confirmation updates.

Finally, the system has minor scalability and testing limitations. While it performs well under normal conditions, handling large-scale global traffic would require further optimisation, and some secondary modules still need more extensive automated testing for complete reliability.

CHAPTER -10 FUTURE ENHANCEMENTS

11.1 Planned Features

The NextGen Travel Intelligence and Automation Portal has been designed with a scalable and modular architecture, allowing future enhancements to be integrated smoothly without affecting existing functionalities. While the current system already supports core features such as travel booking, approvals, payments, and notifications, several advanced improvements are planned to make the platform more intelligent, efficient, and globally adaptable.

One of the key planned features is the integration of AI-based travel intelligence. This will enhance the current smart suggestion system by analysing user travel history, preferences, and booking patterns to generate highly personalised recommendations. The system will also be able to predict price changes for flights and hotels, helping users book at the most cost-effective time.

Another important enhancement is the introduction of a multilingual and multi-currency support system. This will allow users from different regions to access the platform in their preferred language and currency, making the system more user-friendly for international corporate travel management.

A virtual travel assistant is also planned, which will allow users to interact with the system using natural language commands. Employees and managers will be able to request travel bookings, check approval status, and receive updates through a chatbot-style interface, improving ease of use and reducing manual navigation.

To improve transparency and trust, future updates may include advanced analytics and reporting dashboards. These dashboards will provide detailed insights into employee travel patterns, expenses, policy violations, and budget utilisation, helping organisations make data-driven decisions and optimise travel costs.

Another planned enhancement is mobile application support for Android and iOS platforms. This will allow users to manage travel requests, approvals, and notifications on the go, with features like push notifications for approvals, flight updates, and emergency alerts.

Finally, the system aims to include stronger automation and integration capabilities in future versions. This includes automatic policy enforcement using AI rules, better integration with corporate ERP systems, and improved real-time synchronization between booking, approval,

and expense modules to create a fully automated enterprise travel management ecosystem.

11.2 Possible Integrations or Optimisations

The NextGen Travel Intelligence and Automation Portal can be further enhanced through integration with advanced external systems such as airline APIs, railway services, and hotel aggregators. Future integration with global travel APIs like Amadeus or similar platforms can improve data accuracy, expand travel options, and provide real-time availability with better reliability.

Another important optimisation is the improvement of caching and data handling mechanisms. By implementing advanced caching strategies using tools like Redis and introducing smarter data refresh cycles, the system can reduce API calls, improve response time, and ensure faster loading of travel search results even during high traffic.

The platform can also benefit from adopting a microservices architecture in future updates. By separating modules such as booking, payments, notifications, and recommendations into independent services, the system can achieve better scalability, easier maintenance, and improved fault tolerance during high usage conditions.

To improve user experience, future versions can include Progressive Web App (PWA) support or a dedicated mobile application. This will allow users to access the system seamlessly on smartphones with offline capabilities, push notifications, and faster navigation compared to the current web-based version.

Another key optimisation is enhancing the recommendation system using machine learning techniques. By analysing user travel history, preferences, and behaviour patterns, the system can generate more accurate and personalised travel suggestions, improving decision-making and user satisfaction.

Finally, performance and scalability can be further improved through cloud-based optimisations such as load balancing, CDN integration, and auto-scaling services. These enhancements will ensure the platform remains stable, responsive, and efficient even when handling large numbers of users and concurrent booking requests.

CHAPTER -11 CONCLUSION

12.1 Summary of the Project

The NextGen Travel Intelligence and Automation Portal is a comprehensive web-based system designed to simplify and modernize corporate travel management. It integrates travel planning, booking, approvals, and expense tracking into a single unified platform, reducing the complexity of handling multiple disconnected systems.

The primary objective of the project is to automate corporate travel workflows and eliminate manual intervention in processes such as travel requests, managerial approvals, and booking confirmations. This ensures faster processing, improved accuracy, and better compliance with organizational travel policies.

The system is designed to support end-to-end travel operations, including flight booking, hotel reservations, and itinerary management. It allows employees to submit travel requests while enabling managers to review and approve them in real time through a centralized dashboard.

From a technical perspective, the platform follows a full-stack web architecture using modern technologies. The frontend is built using React.js, providing a responsive and interactive user interface, while the backend is developed using Node.js and Express.js for handling APIs and business logic.

A key feature of the project is its role-based access control system. It differentiates between employees, managers, and administrators, ensuring that each user type has specific permissions and access rights based on their role within the organization.

Security is a major focus of the platform, implemented using JWT-based authentication and encrypted communication channels. Passwords are securely hashed, and tokens are used to maintain stateless sessions, improving both security and scalability.

The platform also includes a smart travel suggestion system that recommends flights and hotels based on availability, cost, and company policy constraints. This helps users make informed decisions quickly without manually comparing multiple options.

The approval workflow module ensures that all travel requests follow a structured hierarchy. Employees submit requests, managers approve or reject them, and the system automatically updates the booking status, ensuring transparency at every stage.

Another important component is the expense tracking and billing module, which records all travel-related costs and generates structured summaries. This helps organizations monitor travel budgets and maintain financial control.

The backend architecture is designed to be scalable and modular, allowing easy addition of new services in the future. Each module such as booking, authentication, and notifications operates independently while communicating through APIs.

Database management is handled using structured schemas that store user data, travel requests, booking records, and approval logs. This ensures data consistency, easy retrieval, and efficient reporting.

The frontend design focuses on simplicity and usability, providing intuitive dashboards for both employees and managers. Users can easily navigate through travel requests, booking options, and approval sections without technical complexity.

The system is optimized for performance using caching mechanisms and efficient API handling. This ensures that even during high usage, the platform maintains fast response times and smooth user experience.

The platform also includes analytics capabilities that help organizations understand travel patterns, expenses, and employee behavior. These insights support better decision-making and cost optimization.

Overall, the NextGen Travel Intelligence and Automation Portal successfully demonstrates the integration of modern web technologies to solve real-world corporate booking challenges. It improves efficiency, reduces manual workload, and enhances transparency.

In conclusion, the project represents a complete, scalable, and intelligent travel management solution that combines automation, security, and usability. It is designed not only as an academic project but also as a practical system capable of real-world deployment and future expansion.

12.2 What Was Achieved

NextGen Travel Intelligence and Automation Portal is one of the core components of the system, designed to provide users with a seamless and efficient way to search, compare, and book flight tickets. It acts as the primary entry point for air travel planning within the platform and is fully integrated with other modules such as hotel booking and itinerary management.

This module allows users to input essential travel details such as departure location, destination, travel date, number of passengers, and preferred travel class. These inputs are processed through the system to generate relevant flight options based on availability and user requirements.

Once the user submits the search criteria, the system fetches real-time flight data from external APIs. This ensures that all flight listings displayed are up-to-date, accurate, and reflect current pricing and availability conditions.

The frontend interface of the flight booking module is designed to be simple, responsive, and user-friendly. It displays flight results in a structured layout, allowing users to easily compare different options based on price, duration, and airline preference.

Each flight result includes key details such as airline name, departure time, arrival time, total duration, number of stops, and fare. This helps users make informed decisions quickly without navigating through multiple pages.

To enhance user experience, filtering and sorting functionalities are included. Users can refine their search results based on cost, timing, stopovers, or airline preference, making the selection process more efficient and personalised.

After selecting a flight, users are directed to the booking page where they must enter passenger details such as name, age, and identification information. The system validates this data in real time to ensure accuracy and completeness.

Error handling mechanisms are implemented to manage situations such as invalid inputs, API failures, or unavailable flights. This ensures that the system remains stable even under unexpected conditions.

Security is an important aspect of the flight booking process. User data is protected using authentication methods and secure API communication channels to prevent unauthorized access or data leakage.

The booking confirmation process also provides a detailed summary of the selected flight, including fare breakdown, passenger details, and travel schedule. This ensures transparency and clarity for the user.

The flight booking workflow is structured in a step-by-step manner, guiding users logically from search to selection, passenger details, and final confirmation. This reduces confusion and improves usability for first-time users.

Overall, the Flight Booking module not only focuses on functionality but also emphasizes performance optimization, scalability, reliability, and future extensibility, making it a robust part of the Integrated Tourism Platform.

12.3 Skills Learned During Development

The development of the NextGen Travel Intelligence and Automation Portal helped us gain strong practical knowledge in React.js frontend development. We improved our understanding of component-based architecture, React Hooks such as `useState` and `useEffect`, and state management for building dynamic user interfaces like flight listings, seat selection, and booking summaries.

We also learned how to effectively use React Router for navigation and data flow between pages. Passing flight details using location state and maintaining a smooth multi-page booking process helped us understand how single-page applications are structured in real-world systems.

A major skill developed was working with dynamic UI logic and interactive components, especially in the seat selection module. We implemented seat maps, handled seat selection and removal, and categorized seats into economy, business, and premium sections, which improved our logical thinking and UI development skills.

We gained experience in handling form data and state synchronization, particularly while managing passenger details. Creating dynamic passenger forms based on selected seats and validating input fields improved our understanding of real-time form handling in React applications.

Another important learning was the use of `localStorage` for data persistence. We stored booking details such as selected seats, passenger information, and PNR numbers, which helped us maintain data across multiple pages without backend dependency.

We also implemented QR code generation for digital tickets, which enhanced our knowledge of data encoding and real-world ticketing systems. This feature demonstrated how booking information can be converted into a scannable digital format for verification.

The project improved our understanding of UI/UX design principles, including layout structuring, color coding, and responsive design. Designing booking cards, seat layouts, and ticket interfaces helped us create user-friendly and visually clear interfaces.

Finally, this project strengthened our overall software development mindset, helping us understand how to build a complete end-to-end system with proper flow, modular structure, and user-focused design.

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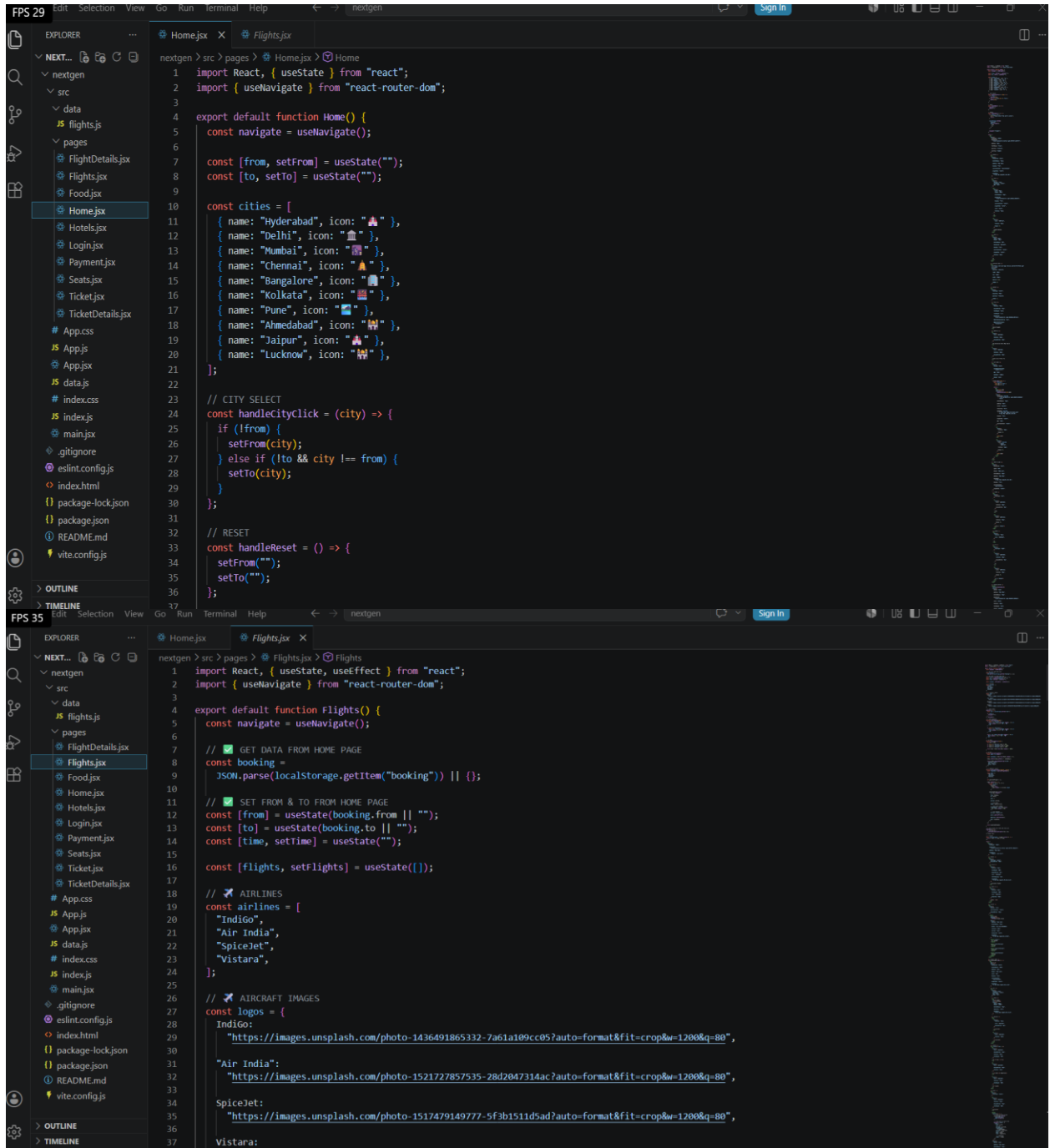
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Sample code snippets



The image displays two screenshots of a VS Code editor interface, showing code snippets for a Next.js project. The top screenshot shows the `Home.jsx` file, and the bottom screenshot shows the `Flights.jsx` file.

Top Screenshot (Home.jsx):

```
1 import React, { useState } from "react";
2 import { useNavigate } from "react-router-dom";
3
4 export default function Home() {
5   const navigate = useNavigate();
6
7   const [from, setFrom] = useState("");
8   const [to, setTo] = useState("");
9
10  const cities = [
11    { name: "Hyderabad", icon: "🏠" },
12    { name: "Delhi", icon: "🏠" },
13    { name: "Mumbai", icon: "🏠" },
14    { name: "Chennai", icon: "🏠" },
15    { name: "Bangalore", icon: "🏠" },
16    { name: "Kolkata", icon: "🏠" },
17    { name: "Pune", icon: "🏠" },
18    { name: "Ahmedabad", icon: "🏠" },
19    { name: "Jaipur", icon: "🏠" },
20    { name: "Lucknow", icon: "🏠" },
21  ];
22
23  // CITY SELECT
24  const handleCityClick = (city) => {
25    if (!from) {
26      setFrom(city);
27    } else if (!to && city !== from) {
28      setTo(city);
29    }
30  };
31
32  // RESET
33  const handleReset = () => {
34    setFrom("");
35    setTo("");
36  };
37
```

Bottom Screenshot (Flights.jsx):

```
1 import React, { useState, useEffect } from "react";
2 import { useNavigate } from "react-router-dom";
3
4 export default function Flights() {
5   const navigate = useNavigate();
6
7   // GET DATA FROM HOME PAGE
8   const booking =
9     JSON.parse(localStorage.getItem("booking")) || {};
10
11  // SET FROM & TO FROM HOME PAGE
12  const [from] = useState(booking.from || "");
13  const [to] = useState(booking.to || "");
14  const [time, setTime] = useState("");
15
16  const [flights, setFlights] = useState([]);
17
18  // AIRLINES
19  const airlines = [
20    "Indigo",
21    "Air India",
22    "SpiceJet",
23    "Vistara",
24  ];
25
26  // AIRCRAFT IMAGES
27  const logos = {
28    Indigo:
29      "https://images.unsplash.com/photo-1436491865332-7a61a109cc05?auto=format&fit=crop&w=1200&q=80",
30    "Air India":
31      "https://images.unsplash.com/photo-1521727857535-28d2047314ac?auto=format&fit=crop&w=1200&q=80",
32    SpiceJet:
33      "https://images.unsplash.com/photo-1517479149777-5f3b1511d5ad?auto=format&fit=crop&w=1200&q=80",
34    Vistara:
35
```

Installation/setup instructions

The NextGen Travel Intelligence and Automation Portal is developed using React.js and Vite to provide a fast, responsive, and user-friendly travel booking experience. Before executing the application, Node.js and npm must be installed on the system to support the project dependencies and runtime environment. The source code is opened in Visual Studio Code, and all required packages are installed using the npm install command. The project utilizes several libraries such as React Router DOM for navigation, QRCode React for ticket QR generation, React Icons for user interface enhancements, and Recharts for analytics visualization. After successful installation of dependencies, the application is executed using the npm run dev command, which launches the development server and generates a local host URL for accessing the platform through a web browser.

Once the application is launched, users are presented with the Login page where authentication details are entered to access the system. After successful login, the user is redirected to the Home page, which serves as the central navigation point for the platform. From there, users can search available flights, select their preferred flight, choose seats, add food preferences, and complete the payment process through the integrated payment module. Upon successful payment, the system automatically generates a digital ticket containing a unique PNR number and QR code for verification purposes. All booking-related information, including passenger details, flight details, selected seats, food orders, payment information, and booking history, is stored securely in the browser's Local Storage for seamless retrieval and management.

The platform further enhances the travel experience by providing hotel recommendations near the destination and a live tracking module that displays real-time flight information such as current status, estimated arrival time, and flight location updates. In addition, the Analytics Dashboard automatically collects booking data after every successful transaction and presents valuable insights such as total bookings, total revenue, seats sold, food orders, and recent booking history. This setup process ensures that all modules of the SkyBook platform function efficiently and are fully integrated, enabling users to experience a complete end-to-end tourism and travel booking solution within a single application.

